LoRA and Privacy: When Random Projections Help (and When They Don't)

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Outline

1. Background & Motivation
2. Problem Definition
3. Method
4. Experiments & Results
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What is Differential Privacy (DP)?

Core Idea

A randomized algorithm $\mathcal{A}$ is (ε, δ)-**differentially private** if for all neighboring datasets $S \sim_H S'$ (differing in one entry) and all measurable E:

$$\Pr[\mathcal{A}(S) \in E] \leq e^{\varepsilon} \Pr[\mathcal{A}(S') \in E] + \delta$$

- **Intuition**: Changing one data point barely affects the output distribution
- ε: privacy budget (smaller $\Rightarrow$ more private)
- δ: failure probability (typically $\ll 1/n$)
- **Gold standard** for protecting training data in ML

The Gaussian Mechanism

ℓ_2 Sensitivity

$$\Delta := \max_{S \sim_H S'} \|f(S) - f(S')\|_2$$

Gaussian Mechanism

For query $f : \mathcal{S} \to \mathbb{R}^d$ with sensitivity Δ, release:

$$\mathcal{A}(S) = f(S) + Z, \quad Z \sim \mathcal{N}(0, \sigma^2 I_d), \quad \sigma \geq \frac{2\Delta\sqrt{\log(1.25/\delta)}}{\varepsilon}$$

This is (ε, δ)-DP.

Key trade-off: More noise $\Rightarrow$ better privacy but worse utility.

The Wishart Projection Mechanism

Definition 1: Projection Mechanism

Given dataset collection $\mathcal{S}$ and query $f : \mathcal{S} \to \mathbb{R}^{n \times d}$, the **Wishart projection mechanism** is:

$$\mathcal{A}_r(S) := f(S) \cdot M, \quad M := Z^\top Z, \quad Z \in \mathbb{R}^{r \times d},\ z_k \sim \mathcal{N}\left(0, \tfrac{1}{r} I_d\right)$$

Equivalently, $M \sim \mathcal{W}_d(\frac{1}{r} I_d, r)$ (Wishart distribution).

Wishart projection is random positive semidefinite

This paper: Analyze $\mathcal{A}_r$ as a proxy for LoRA's built-in randomness.

DP-SGD: Private Training in Practice

Standard SGD:

1. Sample a mini-batch B
2. Per-example gradients $g_i = \nabla_\theta l(\theta; x_i)$
3. Batch gradient $\bar{g} = \frac{1}{|B|} \sum_{i \in B} g_i$
4. Update parameters $\theta \leftarrow \theta - \eta \bar{g}$

DP-SGD [Abadi et al., 2016]:

1. Sample a mini-batch B
2. Compute per-example gradients $g_i = \nabla_\theta l(\theta; x_i)$
3. **Clip** $\tilde{g}_i = g_i \min(1, \frac{C}{\|g_i\|_2})$
4. **Add noise** and compute the batch gradient $\hat{g} = \frac{1}{|B|}(\sum_{i \in B} g_i + N(0, \sigma^2 C^2 I))$
5. Update parameters $\theta \leftarrow \theta - \eta \hat{g}$

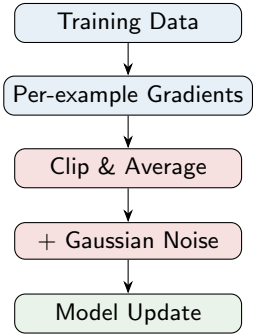
DP-SGD: Private Training in Practice

Standard DP-SGD [Abadi et al., 2016]:

1. Per-example gradient clipping
2. Add Gaussian noise to averaged gradient
3. Privacy accounting via composition

Problems:

- Computationally expensive
- Substantial utility loss for large models
- Noise scales with model dimension d



Fine-tuning Paradigm: Why PEFT Matters

Practical Observation

In many deployments, sensitive data enters only during **fine-tuning**, not pre-training.

Strategy: Start from a public pre-trained model, enforce privacy *only* during fine-tuning.

Parameter-Efficient Fine-Tuning (PEFT):

- Update only a **small subset** of parameters; freeze the rest
- Dramatically reduces memory & compute
- Natural question: Can PEFT also reduce the **cost of DP fine-tuning**?

Low-Rank Adaptation (LoRA)

LoRA [Hu et al., 2022]

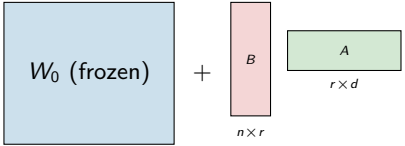
Augment pre-trained weight $W_0 \in \mathbb{R}^{n \times d}$ with a low-rank update:

$$W = W_0 + BA, \quad B \in \mathbb{R}^{n \times r},\ A \in \mathbb{R}^{r \times d},\ r \ll \min\{d, n\}$$

Freeze W_0, only optimize (B, A).

Key properties:

- Trainable params: $r(n + d)$ vs nd
- Often matches full fine-tuning quality
- Inherent **random initialization** of A



LoRA-FA and the Random Projection Structure

LoRA-FA [Zhang et al., 2023]: Fix A (Gaussian initialized), only update B.

Update rule: $B_{t+1} = B_t - \eta(\nabla_W \mathcal{L}(W_t))A^\top$
After T steps:

$$\boxed{W_T = W_0 - \eta \sum_{t=1}^{T} \nabla_W \mathcal{L}(W_t) \cdot \underbrace{(A^\top A)}_{\text{rank-}r\text{ Wishart}}}$$

Key Observation

Each gradient is **right-multiplied** by the random matrix $A^\top A$. This is precisely a **Wishart projection**!

Prior conjecture [Malekmohammadi & Farnadi, 2024]: This randomness may yield DP guarantees.

Research Questions & Contributions

Q1 Is the Wishart projection mechanism DP for **vector-valued** queries?
$\Rightarrow$ Yes (Theorem 1): non-asymptotic DP under alignment condition

Q2 Is the Wishart projection mechanism DP for **matrix-valued** queries (i.e., LoRA)?
$\Rightarrow$ No (Proposition 2): $\varepsilon = \infty$, near-perfect MIA (AUC > 0.99)

Q3 If we add noise, does Wishart randomness **amplify** privacy?
$\Rightarrow$ Yes: strictly better than Gaussian mechanism alone, in both large-r and small-r regimes

Implication for Practice

Low-rank fine-tuning can achieve **smaller** ε than full fine-tuning at the **same noise level**.

Vector Case: Minimum Alignment

For normalized query $f : \mathcal{X}^n \to \mathbb{R}^d$ with $\|f(S)\|_2 = 1$, define:

Minimum Alignment

$$\rho(\mathcal{D}, f) := \min_{\substack{S, S' \in \mathcal{D} \\ S \sim_H S'}} f(S)^\top f(S') \in [-1, 1]$$

Interpretation:

- Large $\rho \approx 1$: neighboring queries are nearly co-directional $\Rightarrow$ hard to distinguish $\Rightarrow$ stronger privacy
- Analogous to "sensitivity" Δ for Gaussian mechanisms

Theorem 1: Vector-Valued DP Guarantee

Theorem 1 (Vector Privacy)

Let $\rho > 0$ be the minimum alignment. For $\delta' > 0$, if $\rho > \frac{t_r(1-\delta')}{\sqrt{r+t_r(1-\delta')^2}}$, the projection mechanism with rank r is $(\varepsilon_\rho, \delta_\rho)$-DP on $\mathcal{D}$:

$$\delta_\rho = \mathbb{E}_{x\sim\chi_r^2}\left[\Phi\left(-\frac{\rho\sqrt{x}}{\sqrt{1-\rho^2}}\right)\right] + 3\delta'$$

$$\varepsilon_\rho \leq \frac{d-r+1}{2}\ln(\rho+K) + \frac{(1-\rho+K)\kappa_{d+r-1}(1-\delta')}{2(\rho-K)}$$

where $K = \sqrt{\frac{1-\rho^2}{r}}\, t_r(1-\delta')$. $t_l(\cdot)$ and $\kappa_l(\cdot)$ denote the quantile functions of the Student-t and χ^2 distributions, respectively, each with l degrees of freedom.

Key insight: Privacy comes from **Wishart randomness alone**—no additive noise needed!

Vector Case: Privacy vs Rank Trade-off

Non-monotonic dependence of ε on rank r:

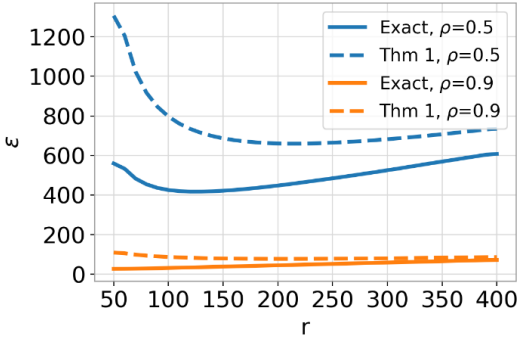
Two competing effects:

- Smaller r: lower-dimensional subspace $\Rightarrow$ compression hides information
- Even smaller r: fewer Gaussian directions $\Rightarrow$ higher tail risk

Optimal r exists that balances both effects.

Figure 1 (from paper):

ε vs r at $\delta = 0.01$, $d = 400$



Negative Result: Matrix Case is NOT Private

Proposition 2 (No Privacy for Matrices)

Let $f : \mathcal{S} \to \mathbb{R}^{n \times d}$ be non-trivial. Then $\exists$ neighboring S, S' and measurable E with:

$$\Pr[\mathcal{A}(S) \in E] = 1, \quad \Pr[\mathcal{A}(S') \in E] = 0$$

$\Rightarrow$ **Not** (ε, δ)-DP for **any** ε and any $\delta < 1$.

Proof intuition:

- Let $V = f(S)$, $V' = f(S')$, $\Delta V = V - V' \neq 0$ with $\text{rank}(\Delta V) = s \geq 1$
- For outputs to coincide: need $(\Delta V)M = 0$, i.e., $(\Delta V)Z = 0$, i.e., every column of Z orthogonal to rowspace of ΔV
- But rowspace has dimension s; orthogonal complement is $(d - s)$-dimensional
- A continuous Gaussian vector lands in a fixed lower-dim subspace with **probability 0**
- $\Rightarrow$ Output supports are **disjoint** almost surely!

Noisy Projection Mechanisms: Setup

Motivation: Wishart randomness alone fails for matrices. Add small Gaussian noise to enforce distributional overlap.

Definition 3: Two Noisy Variants

Given $V \in \mathbb{R}^{d \times n}$, noise $\Xi = [\xi_1, \ldots, \xi_n]$ with $\xi_k \sim \mathcal{N}(0, \sigma_G^2 I_d)$:

M1: $\mathcal{A}_1(V) = MV + \Xi$ (project then add noise)
M2: $\mathcal{A}_2(V) = M(V + \Xi)$ (add noise then project)

- **M1**: Analyzed in the **large**-r regime (Section 4.1)
- **M2**: Analyzed in the **small**-r regime (Section 4.2) — more practical for LoRA

Large-r Regime: Privacy of M1

Challenge: Same M reused across columns $\Rightarrow$ adversary observes $n-1$ columns and learns about M.

Key technique: Orthogonal decomposition $M = M_{\|} + M_{\perp}$

- $M_{\|}$: part in span of observed columns
- $M_{\perp}$: residual, **independent of observed columns** given $H = \text{rowspan}(G)$

For column v_j changed: $Mv_j + \xi_j = \underbrace{(M_{\|} v_j + \xi_j)}_{C(v_j)\text{: correlated}} + \underbrace{M_{\perp} b}_{R(b)\text{: residual}}$

- Correlated term: Gaussian mechanism with high-prob sensitivity bound for $M_{\|}$
- Residual term: Vector-DP analysis (Theorem 1) applies due to **posterior stability**

$\Rightarrow$ Compose: $\varepsilon_{\text{one}} \leq \varepsilon_{\|} + \varepsilon_{\perp}$, $\delta_{\text{one}} \leq \delta_{\|} + \delta_{\perp} + \delta_s$

Large-r: Comparison with Gaussian Baseline

Theorem 3 (Multi-column Composition)

For $V \sim_A V'$ with $k = |J(V, V')|$ differing columns:

$$\varepsilon \leq k\,\varepsilon_{\text{one}}, \quad \delta \leq k\,e^{(k-1)\varepsilon_{\text{one}}}\delta_{\text{one}}$$

Ratio to classical Gaussian mechanism:

$$\frac{\varepsilon^{(k)}_{\text{ours}}}{\varepsilon^{(k)}_{\text{class}}} \leq \sqrt{k}\left[\underbrace{\frac{(\sqrt{d}+\sqrt{p})\sqrt{p}}{(\sqrt{d}+\sqrt{r})^2}}_{\text{Gaussian part}} + \underbrace{\frac{\varepsilon_\perp}{\sigma_M^2(\sqrt{d}+\sqrt{r})^2 K_G}}_{\text{Residual part}}\right]$$

where $p = \text{rank}(U^\top Z) \leq s < r$.

When $s < r$: coefficient $< 1 \Rightarrow$ **our bound strictly improves** on the classical baseline!

Small-r Regime: Privacy of M2

Key insight: Output of $\mathcal{A}_2 = M(V + \Xi)$ lies in $\mathrm{col}(M) = \mathrm{col}(Z)$, a random rank-$r$ subspace.

Conditioned on M, privacy depends only on **projected difference** $P_M \Delta V$:

$$L_M(Y) := \log \frac{dP(\cdot|M)}{dQ(\cdot|M)}(Y) \sim \mathcal{N}\left(-\frac{\|P_M \Delta V\|_F^2}{2\sigma_G^2}, \frac{\|P_M \Delta V\|_F^2}{\sigma_G^2}\right)$$

Define the "good projection" event:

$$\mathcal{G}_\alpha := \left\{M : \|P_M \Delta V\|_F^2 \leq \alpha \|\Delta V\|_F^2\right\}, \quad 0 < \alpha < 1$$

Intuition: A random rank-r subspace captures only $\approx r/d$ of the energy of any fixed direction. When $r \ll d$, most of the sensitivity is "hidden" by the projection.

Theorem 5: Small-r Privacy Bound

Theorem 5 (Privacy of M2)

For $V, V' \in \mathbb{R}^{d\times n}$, $s = \min(n, d)$, any $\alpha \in (0,1)$ and $\varepsilon > 0$:

$$\delta_\alpha(\varepsilon) \le \underbrace{T\left(\varepsilon; \frac{\alpha\|\Delta V\|_F^2}{\sigma_G^2}\right)}_{\text{Gaussian term on } \mathcal{G}_\alpha} + \underbrace{s\left[1 - I_\alpha\left(\frac{r}{2}, \frac{d-r}{2}\right)\right]}_{\text{Failure probability } \Pr(\mathcal{G}_\alpha^c)}$$

where $T(\varepsilon;\mu) = \Phi\left(\frac{-\varepsilon+\mu/2}{\sqrt{\mu}}\right) + 1 - \Phi\left(\frac{\varepsilon+\mu/2}{\sqrt{\mu}}\right)$.

Trade-off in α:

- Smaller $\alpha \Rightarrow$ tighter Gaussian term (effective sensitivity drops)
- Larger $\alpha \Rightarrow$ smaller failure probability
- **Optimize over** α numerically!

Small-r: Why It Beats the Gaussian Baseline

Classical Gaussian (ignores M): $\delta_{\text{Gauss}}(\varepsilon) = T\left(\varepsilon; \frac{\|\Delta V\|_F^2}{\sigma^2}\right)$

Ours: $\delta_{\text{ours}}(\varepsilon) \leq T\left(\varepsilon; \frac{\alpha\|\Delta V\|_F^2}{\sigma^2}\right) + s\left[1 - I_\alpha\left(\frac{r}{2}, \frac{d-r}{2}\right)\right]$

Corollary 1

If $r \ll d$ and $\log s \lesssim r$, then choosing $\alpha \approx r/d$:

$$\delta_{\text{ours}}(\varepsilon) < \delta_{\text{Gauss}}(\varepsilon) \quad \text{(strict improvement)}$$

Why:

- Effective sensitivity drops from $\|\Delta V\|_F$ to $\sqrt{\alpha}\|\Delta V\|_F \approx \sqrt{r/d}\|\Delta V\|_F$
- Beta tail $\Pr(\mathcal{G}_\alpha^c)$ decays exponentially when α is near the typical level r/d
- Net effect: **less noise needed** for same privacy budget

MIA Setup: Threat Model & Evaluation Protocol

Goal: Empirically validate Proposition 2 — can an adversary tell whether a canary was in training?
Target $\mathcal{A}$: 3-layer CNN + 2 FC, **LoRA-FA without noise**. Regimes: scratch on CIFAR-10; pretrain CIFAR-100 $\rightarrow$ fine-tune CIFAR-10.

Membership Inference Attack

Target data $D \subset$ CIFAR-10, $|D| = 5000$. Per trial:

1. Adversary picks canary (x_q, y_q)
2. Sample $b \sim \text{Bernoulli}(1/2)$
3. Train target:
$$f_{\text{target}} \leftarrow \begin{cases} \mathcal{A}(D \cup \{(x_q, y_q)\}) & b{=}1 \\ \mathcal{A}(D) & b{=}0 \end{cases}$$
4. Adversary sees $f_{\text{target}}, D, (x_q, y_q), \mathcal{A}$; outputs $\hat{b}$

Strong adversary knows **all but one** record — DP's worst case.

Canary (static-poison)

- $x_q \sim \mathcal{N}(0, I)$, shape $3{\times}32{\times}32$
- Reference $f = \mathcal{A}(D)$; label $y_q = \arg\min_y [f(x_q)]_y$ (**least-likely** class)

Shadow-Model Evaluation

- $N_{\text{in}}{=}N_{\text{out}}{=}1000$ models
- Score $s = \ell(f, (x_q, y_q))$; lower $\Rightarrow$ member
- Report **ROC-AUC** & best balanced acc.

MIA on Noise-Free LoRA (Table 1)

Setting	**Metric**	r=16	r=32	r=128	r=256	r=512
Train from scratch	Test Acc (%)	57.31	61.76	66.36	67.34	68.36
	AUC (%)	99.86	99.64	100.0	100.0	100.0
Pretrain+ Finetune	Test Acc (%)	75.98	75.63	76.12	76.04	76.45
	AUC (%)	99.81	99.98	100.0	100.0	100.0

Takeaways:

- Near-perfect MIA (AUC ≥ 0.99) across **all ranks and both settings**
- Confirms Proposition 2: noise-free LoRA is **not private**
- Uses strong adversary: knows all but one record, crafts adversarial canary

Noisy Projection: MIA Results (Table 2)

AUC (%) with noise (M2):

Noise	r=16	r=128	r=256	r=512
0.1	78	96	99	100
0.5	56	74	76	86

Test Acc (%):

Noise	r=16	r=128	r=256	r=512
0.1	44.6	54.5	57.9	61.7
0.5	33.1	42.6	46.5	50.9

Observations:

- Larger noise $\Rightarrow$ lower AUC (more private)
- Larger r $\Rightarrow$ higher AUC (less private)
- Consistent with Theorem 5: amplification stronger for **smaller** r
- Privacy–utility trade-off visible

Privacy Budget vs Test Accuracy (Figure 3)

Setup: ResNet-50 (pretrained) + linear head on CIFAR-10, $\delta = 10^{-4}$.

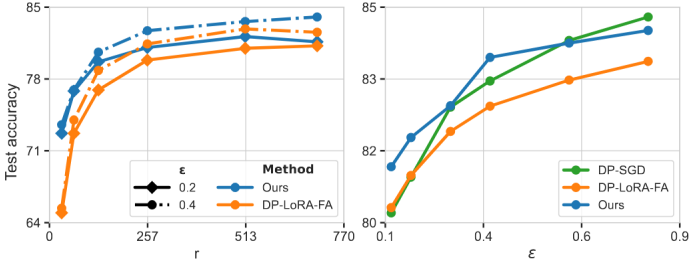


Figure 3 (left) – Accuracy vs rank r:

- M2 consistently $\geq$ DP-LoRA-FA at fixed ε
- Advantage most pronounced at small ranks

Figure 3 (right) – Accuracy vs ε:

- **Small** ε: M2 (Ours) $>$ DP-SGD, because low-rank projection reduces required noise
- **Large** ε: DP-SGD catches up

Connection to LoRA Training

DP-SGD: $\nabla_W \mathcal{L} + \sigma E$	Full-rank, baseline
DP-LoRA-FA: $(\nabla_W \mathcal{L} \cdot A^\top + \sigma E)A$	Standard DP-LoRA
M2 (Ours): $(\nabla_W \mathcal{L} + \sigma' E')A^\top A$	Noise $\rightarrow$ then project

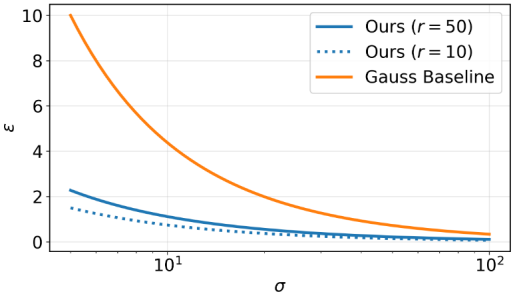
Theoretical Privacy Bound: ε vs Noise Scale (Figure 2)

Figure 2: ε vs σ for $\delta = 10^{-5}$, $d = 2000$.

Observations:

- M2 bound (optimized over α) is strictly below Gaussian baseline
- **Smaller** $r \Rightarrow$ larger gap (more amplification)
- $r = 10$: significant improvement over baseline
- $r = 50$: still better, but gap smaller

Confirms: random projection amplifies privacy, especially in the small-r regime.



Limitations

1. **Data-dependent alignment**: Large-r guarantees (Sec 4.1) rely on alignment parameters $(\rho_{\|}, \rho_{\perp}, \beta_{\perp})$ that are hard to verify in practice
2. **Worst-case adversary**: MIA uses a strong adversary with canary injection; weaker adversaries may see less leakage even without noise
3. **Preliminary experiments**: Only CIFAR-10 + ResNet-50 linear head; large-scale LLM experiments missing

Summary

Setting	Private?	Result
Vector, no noise	✓	Theorem 1
Matrix, no noise	×	Proposition 2 (AUC > 0.99)
Matrix + noise (large r)	✓	Theorem 3–4 (amplification)
Matrix + noise (small r)	✓	Theorem 5 (amplification)

Key Messages

1. LoRA is **not inherently private** despite its random initialization
2. But LoRA's randomness **amplifies privacy** when combined with noise
3. Low-rank fine-tuning can achieve **better privacy–utility trade-off** than full fine-tuning

Q & A

Thank you!

Questions & Discussion

Paper: arXiv:2601.21719